LA TROBE UNIVERSITY Handbook

Course

Master of Information and Communication Technology - Melbourne (Bundoora)

SMINCT | 195 credit points

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Overview

La Trobe's Master of Information and Communication Technology has been carefully designed by industry experts and prepares you for a successful career in the fast-growing information and communication technology (ICT) industry.
Dive straight into the booming world of ICT. Start by honing your fundamental skills in network engineering, programming, web and network security and entrepreneurship in information technology. Then, focus on more specialised topics like campus network design, wireless networking engineering and project management.
Build out your expertise with a mix of electives, from artificial intelligence to virtual reality to Internet of Things programming. Our supportive staff can help you tailor your degree to your career goals.
Put your new knowledge into practice. You'll have the option of applying your skills to either an industry project – alongside one of our industry partners – or a research project in your final year. Get an edge with a CISCO CCNA certification as part of your studies (required for many networking jobs around the world) and complete other industry certifications through our industry partnerships.
From day one you'll have access to our on-campus specialist labs, where you can practise your networking and software development skills using industry-standard facilities.
Throughout your degree, you'll develop skills in entrepreneurship, professional and ethical practices and project management – transferable skills you can apply to roles in any industry.
You'll learn:

- Programming
  - Develop your skills in programming languages and gain insight into areas such as compiler and pre-processor, functions and programming structures, arrays, structures, input/output and the UNIX interface.
- CISCO gears

    - Learn to customise CISCO gears like routers, switches, firewalls and wireless controllers. Use them to provide scalable solutions for corporate communications infrastructure.
- **Project management**
    - Manage projects, communicate effectively and support large teams.
- **Entrepreneurship**
    - Equip yourself with the fundamentals of innovation, professionalism and ethical practices, and gain a competitive edge in the modern business environment.
- **Network system design**
    - Understand how to design and develop large corporate networks, including last-mile wireless segment and network capacity planning. Learn how to secure networks using innovative firewalls and ICT policies.

Portfolio
Office of the Provost

School
Computing, Engineering and Mathematical Sciences

Course coordinator
Lydia Cui

Location(s)
Melbourne (Bundoora)

CRICOS code
061684F

Course instance contact
Lydia Cui

Course duration (full time)
2 years

Course duration (part time)
4 years

Study level
Postgraduate

AQF level
Level 9 - Masters Degree

Available only as an exit award

No

Study options

Domestic

Study mode: Full time, Part time

International onshore

Study mode: Full time

Face To Face: All subjects delivered in person at a specific location.

Multi-Modal: Subjects delivered using a combination of Face To Face, Online and Online Asynchronous.

Online: All subjects delivered Online without the requirement to attend a specific location.

Online Asynchronous: All subjects delivered Online without the requirement to attend at a specific time.

Online plus placement/ intensive: All subjects delivered Online with the requirement of Face To Face attendance for specific intensive (field/industry placement).

Completion requirements

To qualify for the award of Master of Information and Communication Technology, students must complete 195 credit points consisting of:
Industry project pathway
135 core credit-points and
30 core-choice credit points and
30 elective credit points
or
Research Thesis pathway
135 core credit-points and
60 core-choice credit points

Course structure

195 credit points

Level one

90 credit points

Core

90 credit points

LTU0AIM	0 CP	ACADEMIC INTEGRITY MODULE
CSE4001	15 CP	NETWORK ENGINEERING FUNDAMENTALS
CSE5NSW	15 CP	NETWORKS, SYSTEMS AND WEB SECURITY
CSE5002	15 CP	WIRELESS NETWORK ENGINEERING
CSE5003	15 CP	PROFESSIONAL PRACTICES AND ENTREPRENEURSHIP IN INFORMATION TECHNOLOGY
CSE5INE	15 CP	INTERMEDIATE NETWORK ENGINEERING
CSE4IP	15 CP	INTRODUCTION TO PROGRAMMING

Level two

105 credit points

Core

45 credit points

CSE5PM	15 CP	PROJECT MANAGEMENT
CSE5ANE	15 CP	ADVANCED NETWORK ENGINEERING
CSE5FNT	15 CP	FRONTIERS IN NETWORK TECHNOLOGIES

Core choice

60 credit points

Students to select one learning pathway from the list below.

Industry project pathway

60 credit points

Core-choice

15 credit points

Students to select subjects to the value of 15 credit points from the list below.

CSE5IDP	15 CP	INDUSTRY DEVELOPMENT PROJECT

OR

CSE5ARP	15 CP	APPLIED RESEARCH PROJECT

Core-choice

15 credit points

Students to select subjects to the value of 15 credit points from the list below.

CSE5BCC	15 CP	THE BLOCKCHAIN AND CRYPTOCURRENCIES

OR

CSE5CFN	15 CP	COMPUTER FORENSICS

Electives

30 credit points

→ Students to select Level four or Level five CSE coded subjects to the value of 30 credit points.

Research thesis pathway

60 credit points

Core

60 credit points

CSE5001	30 CP	THESIS A
CSE5TSB	30 CP	THESIS B

Course intended learning outcomes

On successful completion you will be able to:

1. Articulate coherently core areas of information and communication technology in the ICT industry, including recent developments and research.
2. Investigate and critically evaluate complex problems using research skills and a mastery of well-integrated information and communication technology knowledge to propose and justify robust solutions
3. Demonstrate how social, legal and ethical issues faced by today's ICT Professional shape and influence the decision-making process
4. Design, evaluate and implement research investigations and software related to the developments of computer and wireless networks, including network security, and theorise about these developments imparting the knowledge to specialist ICT and non-ICT audiences
5. Research and apply independently and collaboratively established theories to the field of ICT, via an applied research team project
6. Demonstrate a high level of personal accountability and autonomy for independent and lifelong learning

Course features

Work based learning (placement) requirements

Elective placement opportunities (Work Based Learning)
During this course, you will have the opportunity to participate in a Work Based Learning (WBL) placement experience, designed to allow you to extend your formal academic learning beyond the classroom. The WBL experience will provide the opportunity to bring your learning from the university into a work environment and test out your knowledge in a professional real-world environment. Some courses include compulsory WBL subjects, however, we also offer a range of elective WBL subjects, both cross-discipline and subject-specific. La Trobe University will source placements for some subjects, while others will require you to source your own placement. If you are required to source your own placement, dedicated staff will guide and support you through the process. Elective WBL subjects generally involve an application process and Subject Coordinator approval. The location and number of hours undertaken can vary considerably depending on the activity and the discipline area.

Work integrated learning opportunities

Industry presentations, Industry informed curriculum, Industry-informed projects, Professional simulations

Career outcomes

ICT specialists are in high demand across the technology, communications, education, finance, government, healthcare and energy sectors.

Possible roles include:

- **Network engineer**
  - Design and use equipment for voice and data. Implement new processes, practices, standards and policies.
- **Network manager**
  - Maintain medium to large-scale corporate networks. Design routing protocols and establish policy frameworks for organisational networks.
- **Senior programmer**
  - Design, develop and test new and existing platforms using local and cloud-based platforms.
- **IT administrator**
  - Design and maintain IT systems within a wide range of organisations and businesses.
- **IT researcher**
  - Conduct research into ICT software, hardware, systems, best practice and trends.

Professional recognition

The Master of Information and Communication Technology is accredited by the Australian Computer Society (ACS). Graduates of the Master of Information and Communication Technology may apply for membership with the ACS. Membership may be subject to additional or ongoing requirements beyond completion of the degree. Please contact the relevant professional body for details.

Other course instances

Select a related course instance to view more information.

SMICTB Bendigo

Master of Information and Communication Technology - Bendigo

Internal course relationships

Replaces

Master of Information and Communication Technology

Credit points: 180

How to apply

StudentOnLine

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